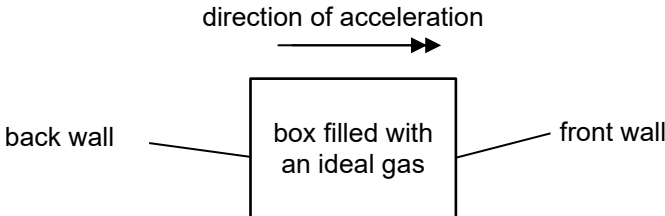


4	(a)	(i)	State what is meant by the <i>internal energy</i> of a system.	
				
				[1]
			Solution: sum of the random kinetic and potential energies of the individual atoms/molecules/particles of the system	1
		(ii)	Explain why, for an ideal gas, the change in internal energy is directly proportional to the change in thermodynamic temperature of the gas.	
				
				[3]
			In an ideal gas, no intermolecular forces so potential energy between atoms/molecules equal to zero. Thus internal energy is solely due to kinetic energy of the atoms/molecules.	1

		(Mean) kinetic energy of atoms/molecules is proportional to the (thermodynamic) temperature of the gas. Therefore the <i>change in</i> internal energy is proportional to the <i>change in</i> thermodynamic temperature of the gas.	1 1
	(iii)	A box, filled with an ideal gas, is rapidly accelerated horizontally from rest. direction of acceleration  Fig. 4.1 Suggest, with a reason, how the internal energy of the gas would vary from the back wall to the front wall of the box during the acceleration. 	
			[2]
		Solution: Region of sudden compression at the BACK → work done on gas → gain energy; Region of sudden expansion at the FRONT → work done by gas → lose energy (Assuming adiabatic process), the internal energy decreases from back towards the front.	1 1
	(b)	A cylinder of volume $1.8 \times 10^4 \text{ cm}^3$ contains helium gas at pressure $6.4 \times 10^6 \text{ Pa}$ and temperature 25°C. The root-mean-square speed of the helium gas atoms is 336 m s^{-1}. Helium gas may be considered to be an ideal monatomic gas.	
	(i)	Calculate the number of helium atoms in the cylinder.	
		number =	
		[2]	
		Solution: $pV = NkT$ $(6.4 \times 10^6)(1.8 \times 10^4 \times 10^{-6}) = N (1.38 \times 10^{-23})(25 + 273.15)$ $N = 2.7999 \times 10^{25} = 2.8 \times 10^{25}$ OR	1 1

			$pV = nRT$ $(6.4 \times 10^6)(1.8 \times 10^4 \times 10^{-6}) = n(8.31)(25 + 273.15)$ $n = 46.496$ $N = n \times N_A$ $= 46.496 \times (6.02 \times 10^{23})$ $= 2.7991 \times 10^{25} = 2.8 \times 10^{25}$	(1)
		(ii)	Calculate the density of helium atoms in the cylinder.	(1)
			density =kg m ⁻³	[2]
			Solution: $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle = \frac{1}{3} \rho \langle c^2 \rangle$ where $\sqrt{\langle c^2 \rangle} = 336$ $(6.4 \times 10^6) = \frac{1}{3} \rho (336^2)$ $\rho = 170.07 = 170 \text{ kg m}^{-3}$	1 1